

Cyber Crime Case Studies

Name: Kanjariya Jeet M.
Enrollment: 241105060063

A detailed analysis of real-world cyber incidents, their impact and key lessons for a safer digital future.

CASE 1 — WANNACRY RANSOMWARE WORM (GLOBAL, 2017)



Illustrative graphic: a WannaCry-style ransom screen. WannaCry locked files on infected Windows computers and demanded payment in bitcoin.

IMAGE DESCRIPTION

The graphic shows the kind of ransom message victims saw on infected computers: files encrypted, a payment demand in bitcoin and a countdown timer meant to create panic. It is an illustration of the incident, not a photograph of a specific victim's screen.

1) NEWS HISTORY — WHAT HAPPENED?

On 12 May 2017 the WannaCry ransomware began spreading across the world. It infected more than 200,000 computers in roughly 150 countries, including hospitals, telecom companies, factories and government offices. In England, the National Health Service was badly affected: appointments were cancelled and some hospitals diverted emergency patients. A security researcher slowed the spread by registering a domain that acted as a kill switch inside the malware. Western governments later attributed the attack to groups linked to North Korea.

SOURCE / NEWS REFERENCE

UK National Audit Office, "Investigation: WannaCry cyber attack and the NHS" (2017); public advisories from Microsoft and national cyber-security agencies, May 2017.

2) ATTACK TECHNIQUE

WannaCry was a ransomware worm. It used an exploit called EternalBlue against a flaw in the Windows SMBv1 file-sharing protocol, for which Microsoft had released a patch (MS17-010) two months earlier. Because it spread from computer to computer by itself, a single unpatched machine on a network could infect many others. It then encrypted files and displayed the ransom note.

3) TOTAL LOSS / FINANCIAL IMPACT

Estimates of the total damage vary widely between sources, but they run into hundreds of millions to billions of US dollars worldwide, mostly from downtime, lost work and recovery rather than ransom payments, which were relatively small. In the NHS, thousands of appointments and operations were cancelled. Many organisations lost access to important records for days.

4) RECOVERY — COMPANY / AUTHORITY RESPONSE

Organisations isolated infected machines, applied the Microsoft patch, restored data from clean backups and rebuilt affected systems. Microsoft released emergency updates even for old, unsupported Windows versions. Governments and agencies issued public warnings, and the NHS later reviewed its cyber-security arrangements.

5) RECOVERY COST AND RECOVERY TIME

The NHS in England reported that around 19,000 appointments were cancelled and put its direct cost at about £92 million, including lost output and IT recovery. Most organisations restored operations within days, but full clean-up and security upgrades took much longer. Global cost figures differ by source and should be quoted with that caveat.

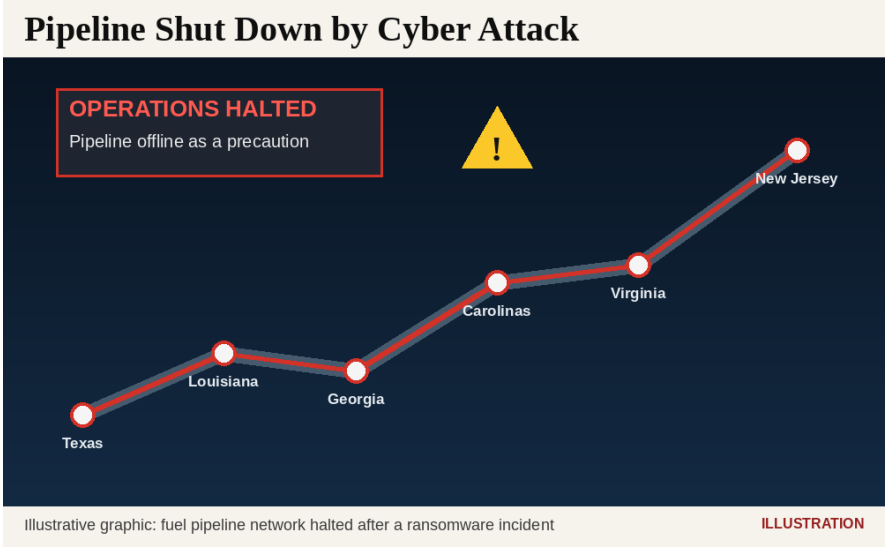
6) HOW COULD IT HAVE BEEN PREVENTED?

Install security patches quickly; disable the old SMBv1 protocol; keep unsupported operating systems off the network or isolated; segment networks so malware cannot spread freely; keep tested offline backups; filter and monitor network traffic; and rehearse an incident-response plan.

SOLUTION / KEY LESSON

A patch that exists but is not installed gives no protection. Fast patching, network segmentation and reliable offline backups together turn a worldwide emergency into a manageable incident.

CASE 2 — COLONIAL PIPELINE RANSOMWARE ATTACK (USA, 2021)



Illustrative graphic: a fuel pipeline network shut down after a ransomware attack on its business systems.

IMAGE DESCRIPTION

The graphic shows a long fuel pipeline running along the U.S. East Coast with a warning that operations were halted. It illustrates how a cyber attack on company IT systems can disrupt a physical service that many people depend on.

1) NEWS HISTORY — WHAT HAPPENED?

On 7 May 2021 Colonial Pipeline, which carries a large share of the fuel used on the U.S. East Coast, discovered a ransomware attack and shut down its pipeline operations as a precaution. The criminal group DarkSide, which sells ransomware to affiliates, was blamed by the U.S. government. Fuel shortages and panic buying followed in several states. The company restarted the pipeline about five to six days later.

SOURCE / NEWS REFERENCE

U.S. Department of Justice announcement on recovery of ransom payment (June 2021); Colonial Pipeline statements and congressional testimony, May–June 2021.

2) ATTACK TECHNIQUE

The attackers gained entry through a single compromised password for a legacy VPN account that, according to later testimony, did not use multi-factor authentication. They then stole data and deployed ransomware on the company's business systems. The pipeline's operational technology was not reported to have been encrypted, but the company shut it down to be safe.

3) TOTAL LOSS / FINANCIAL IMPACT

Fuel supply was disrupted across several states, prices rose and many filling stations ran out of fuel. The company paid a ransom of about 75 bitcoin, roughly 4.4 million US dollars at the time, and also lost revenue during the shutdown. The incident was treated as a national-security issue.

4) RECOVERY — COMPANY / AUTHORITY RESPONSE

Colonial hired outside cyber-security firms, informed the FBI and other agencies, and restored its systems in stages. The U.S. Department of Justice later announced that it had recovered a large part of the ransom by seizing the cryptocurrency wallet. The government issued emergency fuel-transport rules and new security directives for pipeline operators.

5) RECOVERY COST AND RECOVERY TIME

The ransom paid was about 4.4 million US dollars, and the U.S. Department of Justice announced recovery of about 2.3 million dollars of it. The main recovery work took about a week for the pipeline to restart, but investigation, system rebuilding and security upgrades continued for much longer. Total cost to the company and the economy is not reliably known.

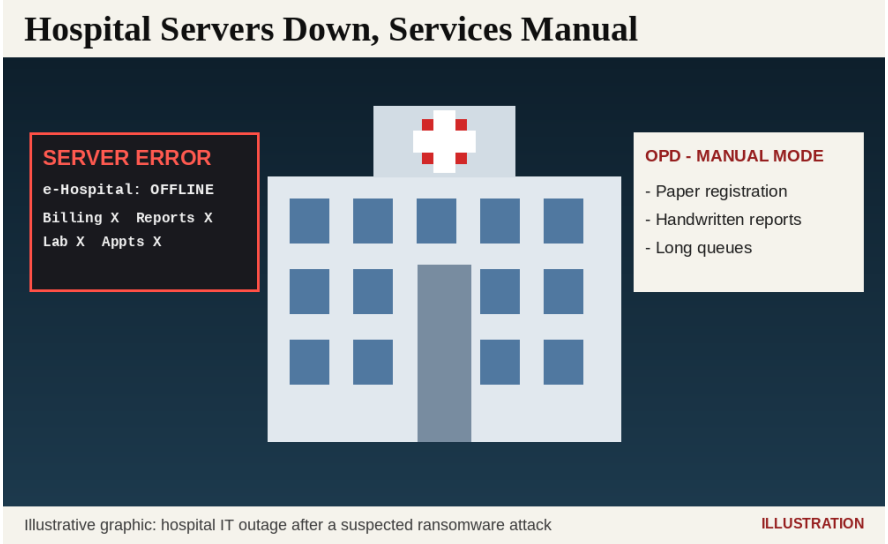
6) HOW COULD IT HAVE BEEN PREVENTED?

Require multi-factor authentication on all remote access; remove unused accounts; keep business and operational networks separated; monitor for unusual logins and data transfers; use unique passwords checked against known breaches; maintain tested backups; and practise shutdown and restart plans.

SOLUTION / KEY LESSON

One weak password on one old account was enough to disrupt critical infrastructure. Multi-factor authentication, network separation and good incident planning are basic protections for essential services.

CASE 3 — AIIMS DELHI RANSOMWARE ATTACK (INDIA, 2022)



Illustrative graphic: a hospital whose e-Hospital servers went offline, forcing staff to work in manual mode.

IMAGE DESCRIPTION

The graphic shows a hospital with its billing, reports, lab and appointment systems offline and a note that outpatient services were running manually. It illustrates the effect of a suspected ransomware attack on a large public hospital.

1) NEWS HISTORY — WHAT HAPPENED?

On 23 November 2022 the servers of the e-Hospital system used at the All India Institute of Medical Sciences (AIIMS), New Delhi went down from about 7 a.m., affecting outpatient registration, billing, report generation and sample collection. The National Informatics Centre identified it as a suspected ransomware attack. Services ran in manual mode with long queues while Delhi Police, CERT-In and other agencies investigated. A case of extortion and cyber terrorism was registered on 25 November 2022.

SOURCE / NEWS REFERENCE

News reports of 23–29 November 2022, including Business Today, Deccan Herald (PTI) and Onmanorama (IANS). Details such as the ransom demand were reported from unnamed sources and are not officially confirmed.

2) ATTACK TECHNIQUE

The attack was described as ransomware: files on affected servers had their extensions changed, which indicated encryption. Early reporting pointed to weaknesses such as network configuration and unpatched or poorly segmented systems, but the exact entry method was not publicly confirmed. Some reports said a ransom of around 200 crore rupees in cryptocurrency was demanded, though other reports said police had no ransom demand on record, so this point is treated as unverified.

3) TOTAL LOSS / FINANCIAL IMPACT

Patient services were disrupted for days: appointments, lab reports and billing had to be handled on paper, and sample collection slowed sharply. Sensitive health data of a very large number of patients was at risk. Two system analysts were reported to have been suspended during the investigation.

4) RECOVERY — COMPANY / AUTHORITY RESPONSE

The National Informatics Centre and AIIMS isolated and cleaned the infected servers, restored data from backup servers, and brought services back gradually while running hospital operations manually. Staff computers connected to the network were reportedly formatted as a precaution. Multiple government agencies investigated the attack.

5) RECOVERY COST AND RECOVERY TIME

No official figure for total recovery cost has been established in the reports used here. Recovery took days to weeks, depending on the service, and manual operation continued during that time. The real cost includes staff effort, delayed treatment and the loss of public trust in the system.

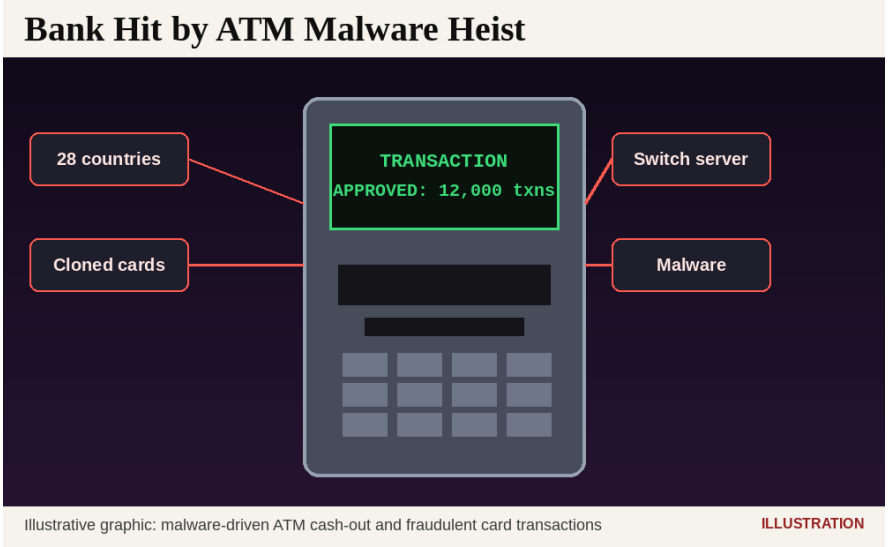
6) HOW COULD IT HAVE BEEN PREVENTED?

Segment hospital networks from the internet and from each other; patch servers and firewalls regularly; use strong access control and multi-factor authentication; keep offline, tested backups; monitor networks continuously; train staff against phishing; and prepare a clear downtime and recovery plan for medical services.

SOLUTION / KEY LESSON

Hospitals are critical services, so a cyber attack becomes a patient-safety issue. Strong backups, network separation and a practised manual fallback plan are essential.

CASE 4 — COSMOS BANK ATM MALWARE ATTACK (INDIA, 2018)



Illustrative graphic: malware on a bank's payment systems used to approve fraudulent ATM withdrawals made with cloned cards.

IMAGE DESCRIPTION

The graphic shows an ATM with labels for malware on the bank's switch server, cloned cards and withdrawals in many countries. It illustrates a coordinated cash-out attack that combined technical compromise with card fraud.

1) NEWS HISTORY — WHAT HAPPENED?

In August 2018 the Pune-based Cosmos Co-operative Bank reported a large cyber attack. Media reports said that on 11 August, fraudulent withdrawals using cloned debit cards were made from ATMs in many countries, followed by a fraudulent SWIFT transfer on 13 August. About 94 crore rupees were reported stolen in total. The case was investigated by the Pune cyber police.

SOURCE / NEWS REFERENCE

News reports, August–September 2018, including statements by Cosmos Bank and Pune police. Some technical details were reported from police sources and are described here with caution.

2) ATTACK TECHNIQUE

Reports said malware was placed on the bank's ATM switch server, which sits between the ATM network and the bank's core systems. The malware approved fraudulent transactions made with cloned Visa and RuPay debit cards, without the bank's normal authorisation checks. In parallel, a SWIFT payment request was used to move money to an overseas account.

3) TOTAL LOSS / FINANCIAL IMPACT

Reported losses were about 94 crore rupees: roughly 78 crore through thousands of ATM withdrawals in many countries, about 2.5 crore through domestic transactions, and about 13.9 crore through the SWIFT transfer. The case showed that even a co-operative bank can be a target of an international attack.

4) RECOVERY — COMPANY / AUTHORITY RESPONSE

The bank shut down its ATM switch and card operations, informed the police and authorities and worked with the card networks to block and trace transactions. Law-enforcement agencies tracked the transfers; the bank later announced that it would restore services and that customers' deposits were safe.

5) RECOVERY COST AND RECOVERY TIME

The headline loss was about 94 crore rupees. The exact net loss after recoveries and any insurance is not clearly established in public reports. Service restoration and security reviews took weeks, and the bank also faced reputational damage.

6) HOW COULD IT HAVE BEEN PREVENTED?

Isolate and monitor ATM switch and payment servers; use application allow-listing and file-integrity monitoring; enforce strict authorisation checks for all card transactions; apply multi-factor approval for SWIFT messages; set limits and alerts for unusual volumes; and run regular security audits of third-party systems.

SOLUTION / KEY LESSON

Payment infrastructure needs layered controls: no single server should be able to approve large sums on its own. Real-time monitoring and strict transaction limits can stop a cash-out attack early.

CASE 5 — TWITTER ACCOUNT TAKEOVER AND BITCOIN SCAM (USA, 2020)



Illustrative graphic: a hijacked verified account promoting a fake cryptocurrency giveaway. The profile shown is a generic placeholder.

IMAGE DESCRIPTION

The graphic shows a hijacked verified-style account posting a fake giveaway that promises to double any bitcoin sent. It illustrates how trusted accounts can be abused for fraud when an attacker gains control of them.

1) NEWS HISTORY — WHAT HAPPENED?

On 15 July 2020 the accounts of many well-known people and companies on Twitter posted messages that asked followers to send bitcoin and promised to return double the amount. Twitter said that attackers had targeted its employees with a phone-based social-engineering attack to obtain access to internal tools. Around 130 accounts were targeted. Arrests in the United States followed within weeks.

SOURCE / NEWS REFERENCE

Twitter (now X) security-incident updates, July–August 2020; U.S. Department of Justice announcement of charges, 31 July 2020.

2) ATTACK TECHNIQUE

The main technique was social engineering, specifically phone spear-phishing (vishing). Attackers pretended to be IT staff to trick employees into giving credentials, then used internal admin tools to reset account access. They posted the scam messages from the hijacked accounts. No software vulnerability was needed.

3) TOTAL LOSS / FINANCIAL IMPACT

Scam victims sent bitcoin worth roughly 120,000 US dollars in total, according to public reports. The wider damage was loss of trust in the platform, which temporarily limited verified accounts from tweeting, and an investigation by regulators and law enforcement.

4) RECOVERY — COMPANY / AUTHORITY RESPONSE

Twitter locked affected accounts, restricted internal tools, and investigated with the FBI and other agencies. It published updates, reset access and promised stronger staff security. U.S. prosecutors charged three people, including a teenager, with offences related to the attack.

5) RECOVERY COST AND RECOVERY TIME

The direct theft was about 120,000 US dollars. Twitter’s total cost, including investigation, regulatory attention and security upgrades, has not been publicly given as a single reliable figure. The incident was resolved within hours, but trust and policy changes continued for months.

6) HOW COULD IT HAVE BEEN PREVENTED?

Train staff to recognise phone-based social engineering; require phishing-resistant multi-factor authentication such as hardware keys; limit who can use powerful admin tools; require approval from a second person for sensitive actions; log and monitor all admin activity; and verify caller identity through a separate channel.

> **SOLUTION / KEY LESSON**
>
> People are often the easiest way into a system. Technical controls like security keys and dual approval must be combined with staff awareness training.